07-MANUFACTURING-100

100-Unit Pilot Run Plan (PCBWay-centric)

Flow

1. EVT: 5 boards, PCBWay rigid-flex + PCBA; 5 CNC shell sets (prototype, before polish).
2. DVT: 20 units; freeze Gerber/BOM/STEP.
3. PVT: 100 PCBA (order 110 for 10 % scrap), 105 shell sets, 110 glass, 110 cells.
4. Assembly: in-house or PCBWay assembly of final product (ask) — pogo flash → `lot-link` QA → seal → mirror clean → pack.

Estimated cost (unverified, USD)

Item	Cost
PCB+PCBA ×110 (incl. stencil, setup)	3,500–5,500
Stainless shells ×105 sets, incl. polish	9,000–13,000
Glass, gasket, LRA, battery, misc	2,500–3,500
Test jig + pogo fixtures	500–1,000
Prototypes/NRE (EVT, DVT)	2,000–3,500
Certification (FCC test, UN38.3 docs)	5,000–8,000
Packaging, shipping, duties	800–1,500
Total	**≈ 23–36k all-in** (≈ 16.5–23.5k excl. NRE & cert)

QA gates per unit

Visual → flash+pair → sensor self-test → Wi-Fi RSSI → button → camera frame → Qi charge 5 min → leak/gap check → serial label.

Traceability

Serial = `LOT-P1-YYMM-NNN`; stored with token fingerprint, fw version, QA CSV.